Measuring Electricity Name _____

1. Match up terms with definitions by writing the correct letter of each definition in the box beside each term.

Definition

a. A complete conducting pathway
b. Meter used to measure the size of a current
c. Negatively charged particles which are parts of atoms
d. Objects which are part of electrical circuits
e. Circuit in which there are two or more pathways
f. The flow of electrons around a conducting pathway
g. Unit of voltage gain or loss
h. Object which converts electrical energy into other forms
i. Particles that all matter is made of
j. Meter used to measure the gain or loss of voltage
k. Unit of current
l. An object that supplies the electrical energy to a circuit
m. Circuit in which all components are in a single loop
n. Part of a circuit that slows the flow of electrons
o. The loss or gain of energy by the current

Terms

	Attempt	
	1st	2nd
Electrons		
Atoms		
Current		
Circuit		
Cell		
Components		
Series circuit		
Parallel circuit		
Ammeter		
Ampere or amp		
Voltage gain or loss		
Volt		
Voltmeter		
Resistor		
Load		

Measuring Electricity Name _____

2. Draw a circuit diagram with:

a) a cell, bulb and ammeter in series

b) a cell, one bulb and a resistor, and another bulb and a voltmeter in parallel.

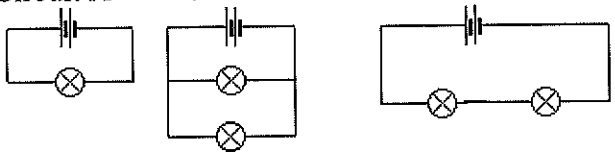
3. Complete.

In a series circuit, the _____is the same at all points. When lamps are connected in series, the voltage gained from the _____ is shared between them so the bulbs will be _____.

In a parallel circuit, the _____ of the currents in the branches is equal to the total _____ flowing in the circuit. When lamps are connected in parallel each bulb receives the full _____ supplied to the current by the batteries, so the bulbs will be _____.

4. The lamps are all of equal strength and the batteries are all 1.5V.

Circuit A Circuit B Circuit C



a. How many volts will the current in each circuit receive? _____

b. In which circuit(s) will the bulbs be brightest? __

c. In which circuit(s) will the bulbs be dimmest? __

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1. Match up terms with definitions by writing the correct letter of each definition in the box beside each term.

Definition

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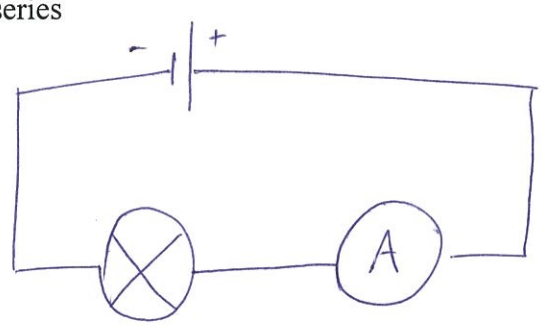
Terms

	Attempt	
	1st	2nd
Electrons	c	
Atoms	i	
Current	f	
Circuit	a	
Cell	l	
Components	d	
Series circuit	m	
Parallel circuit	e	
Ammeter	b	
Ampere or amp	k	
Voltage gain or loss	n	
Volt	g	
Voltmeter	j	
Resistor	o	
Load	h	

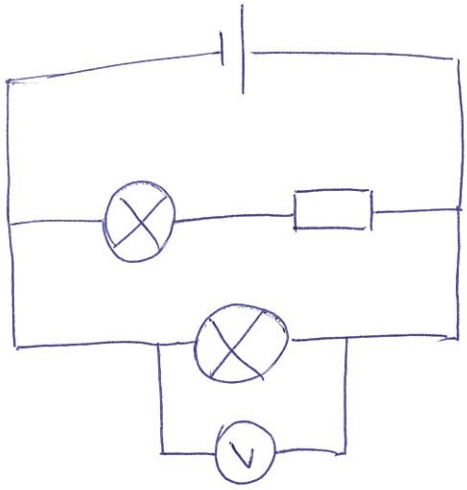
Measuring Electricity Name _____

2. Draw a circuit diagram with:

a) a cell, bulb and ammeter in series



b) one bulb and a resistor, and another bulb and a voltmeter in parallel.



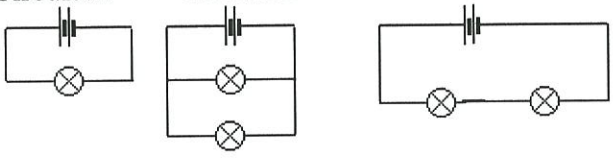
3. Complete.

In a series circuit, the current is the same at all points. When lamps are connected in series, the voltage gained from the cell is shared between them so the bulbs will be dim.

In a parallel circuit, the sum of the currents in the branches is equal to the total current flowing in the circuit. When lamps are connected in parallel each bulb receives the full voltage supplied to the current by the batteries, so the bulbs will be bright.

4. The lamps are all of equal strength and the batteries are all 1.5V.

Circuit A Circuit B Circuit C



a. How many volts will the current in each circuit receive? 1.5 V

b. In which circuit(s) will the bulbs be brightest? A, B

c. In which circuit(s) will the bulbs be dimmest? C

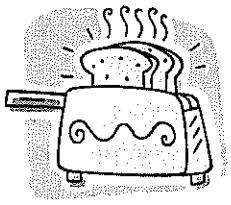
Worksheet: Methods of Heat Transfer
(conduction, convection, and radiation)

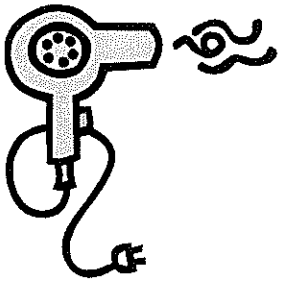
Define conduction:

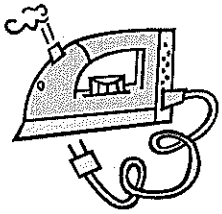
Define convection:

Define radiation:

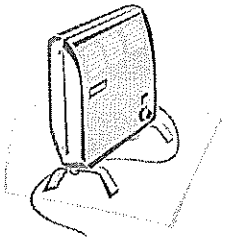
Identify the method of heat transfer that takes place in each illustration. Some illustrations may show more than one form of heat transfer.

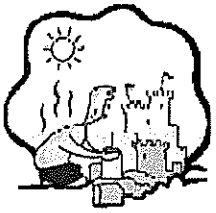
1. 

2. 

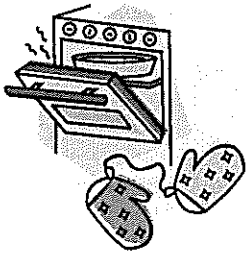
3. 

4. 

5. 

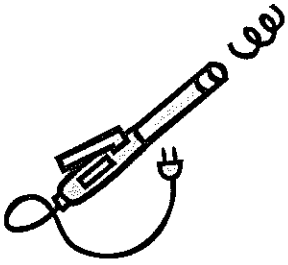
6. 

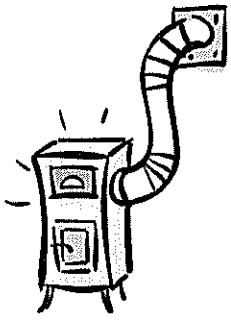
7. 

8. 

9. 

10. 

11. 

12. 

In each of the following situations, identify the method of heat transfer taking place (conduction, convection, radiation). More than one process may be occurring.

1. Hot coffee is stirred with a spoon, the spoon gets hot due to _____.
2. A chair is placed several feet from a fire in a fireplace. The fireplace has a glass screen. The side of the chair facing the fireplace gets warm because of_____.
3. A certain type of decorative lamp contains colored liquids. These liquids form globs that break off and rise to the top of the liquid. The globs rise due to _____.
4. Near the ceiling of a room the air is warmer. The warm air rises because of _____.
5. A college student holds the back of his hand near an iron to see if it is hot. Heat is transferred to his hand by _____.
6. A heater is placed under one corner of a water bed mattress. Warm water moves throughout the mattress because of _____.
7. A certain type of stainless steel cookware has a layer of copper applied to the bottom to help it heat evenly. The copper transfers heat to the pan by _____.
8. In a swimming pool, the water near the surface is slightly warmer. The warm water rises because of _____.
9. One end of a copper rod is placed in a flame of a Bunsen burner. Small pieces of wax placed along the rod melt at progressively larger distance from the flame. Heat is transferred through the rod by _____.
10. A house burns down. On the house across the street, all of the vinyl siding is twisted and warped by the heat. The heat was transferred across the street by _____.
11. Warm air over the beach rises while cooler dense air from the ocean rushes in due to _____.
12. The metal skewer gets so hot that you drop your marshmallow in the campfire because of _____.
13. A huge rock at the state park gets so hot during the day that you can't sit on it from _____.
14. You lay on that same rock at night so that you can keep warm by _____.
15. A fireman feels a door and it is hot from the fire on the other side due to _____.
16. The cause of weather systems on earth is _____.
17. You are in the top bunk of a bunk bed and you want to turn the air conditioner on while your friend on the bottom bunk is fine is caused by _____.

Worksheet: Methods of Heat Transfer
(conduction, convection, and radiation)

Define conduction:

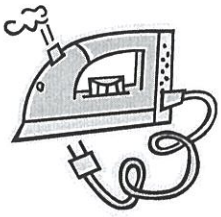
Define convection:

Define radiation:

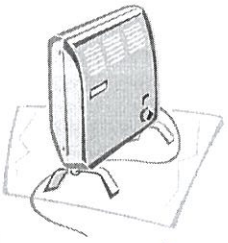
Identify the method of heat transfer that takes place in each illustration. Some illustrations may show more than one form of heat transfer.

1. 
conduction

2. 

3. 
conduction

4. 
radiation

5. 
convection

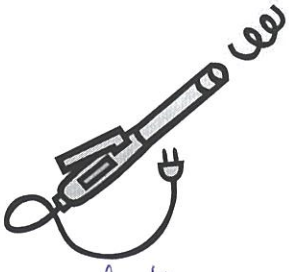
6. 
radiation

7. 
conduction (grilling meat)

8. 
conduction

9. 
convection

10. 

11. 
conduction

12. 
convection

In each of the following situations, identify the method of heat transfer taking place (conduction, convection, radiation). More than one process may be occurring.

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2. A chair is placed several feet from a fire in a fireplace. The fireplace has a glass screen. The side of the chair facing the fireplace gets warm because of radiation.
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